EMF of the cell,

$$E = E^{\circ} - \frac{0.0591}{n} \log \frac{[Zn^{2+}]}{[Ag^{+}]^{2}} \qquad (3.16)^2 = 10$$

$$E = 1.56 - \frac{0.0591}{2} \log \frac{0.1}{10}$$

$$= 1.56 - 0.02955 \times (-2)$$

$$\underline{E = 1.62 \text{ volt}}.$$

Reference Electrode :- The electrodes with known and constant potential which are used to measure the potential of other electrodes are called Ref / 2° ref. electrode

Reference electrodes are the electrodes with reference to those, the electrode potential of any electrode can be measured. standard hydrogen electrode is the primary reference electrode as the electrode potential values of other electrodes are determining using SHE whose potential is arbitrarily taken as zero volts.

Limitations :-

1) construction and working of electrode is difficult on account of difficulties involved in maintaining the concentration at unity & in keeping the pressure of the gas uniformly at one atmosphere.
2) Platinum is highly susceptible for poisoning by the impurities in the gas. (Eg. As)
3) It cannot be used in the presence of oxidising agents.

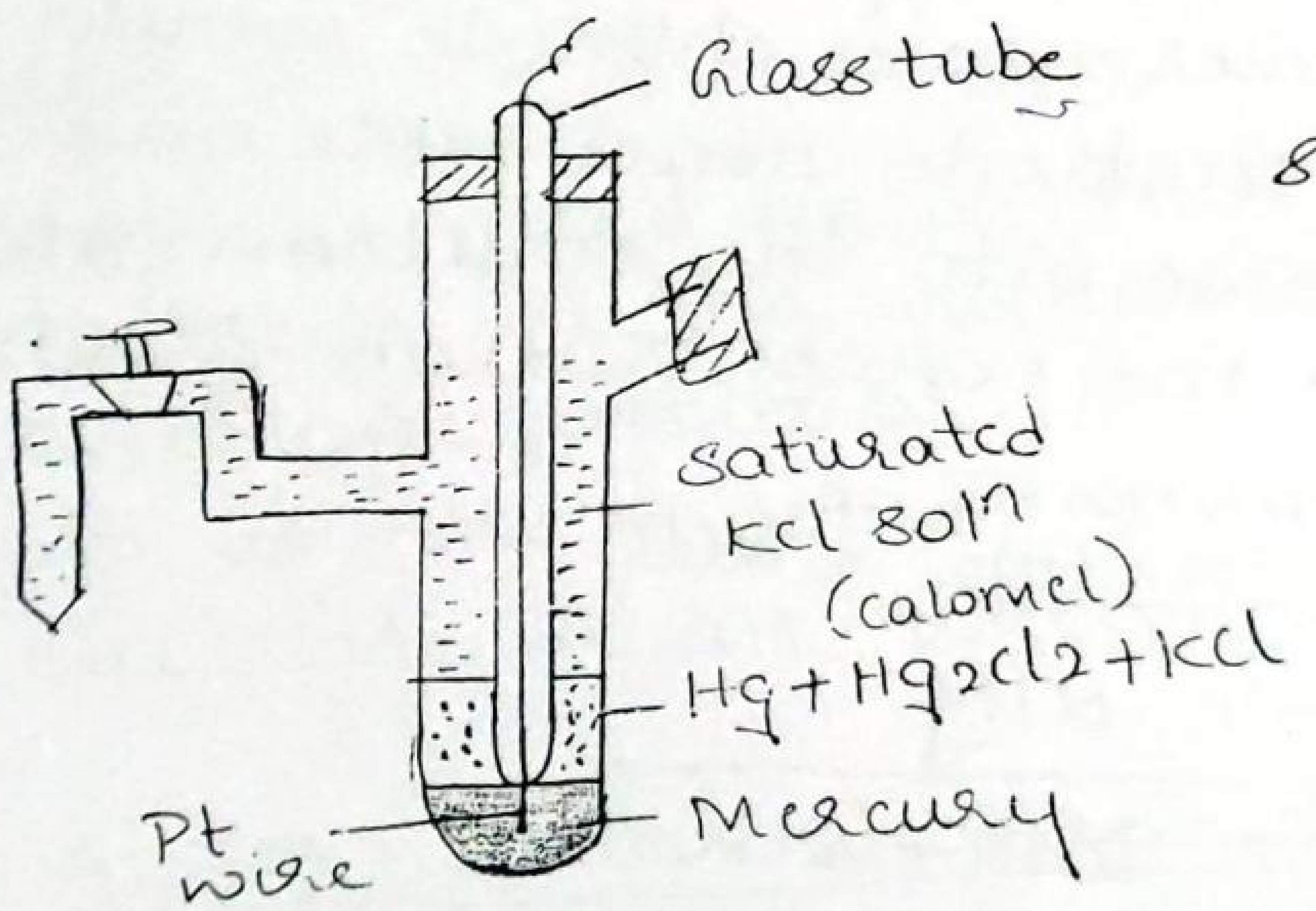
$$Cr_2O_7^{2-} + 14H^+ + e^- \rightarrow 2Cr^{3+} + 7H_2O.$$

Secondary Reference electrode :-

Because of the difficulties involved in the use of standard hydrogen electrode as reference electrode, some other electrodes of constant electrode potential are used refered to as secondary reference electrodes.

Two such electrodes which are in common use are calomel electrode and silver-silverchloride electrode.

(1) Saturated calomel electrode :-



It is a metal-metal salt ion electrode. Calomel electrode consists of mercury, mercurous chloride and a solution of potassium chloride. The electrode is represented as,

$Hg / Hg_2Cl_2 / Cl^-$.

Calomel electrode can be easily set up in the lab. Pure mercury is placed at the bottom of the glass tube having a side tube on each side. Above this a calomel paste obtained by grinding mercury, mercurous chloride and potassium chloride solution is placed. Then a saturated solution of KCl is introduced above the paste through the side tube. A platinum wire which is sealed into the glass tube is dipped into mercury to provide the external electrical contact.

If the electrode reaction involves oxidation, then it would liberate electrons and send Hg_2^{2+} ions into the solution. This Hg_2^{2+} ions combine with Cl^- ions (by KCl) forming sparingly soluble Hg_2Cl_2. The result is a fall in the concentration of Cl^- ions in the solution.

$$2Hg \longrightarrow Hg_2^{2+} + 2e^-$$
$$Hg_2^{2+} + 2Cl^- \longrightarrow Hg_2Cl_2$$
$$\overline{2Hg + 2Cl^- \longrightarrow Hg_2Cl_2 + 2e^-}$$

On the other hand if the electrode reaction involves reduction, then the Hg_2^{2+} ions furnished by the sparingly soluble mercurous chloride would be discharged at the electrode. Hence more and more calomel would pass into the solution. The result is increase in the concentration of Cl^- ions.

$$Hg_2Cl_2 \longrightarrow Hg_2^{2+} + 2Cl^-$$
$$Hg_2^{2+} + 2e^- \longrightarrow 2Hg.$$
$$\overline{Hg_2Cl_2 + 2e^- \longrightarrow 2Hg + 2Cl^-}$$

Thus the net reversible electrode reaction is, $Hg_2Cl_2 + 2e^- \rightleftharpoons 2Hg + 2Cl^-$

The concentration of KCl solution used is either decinormal, normal or saturated solution. Correspondingly the electrode is known as decinormal, normal or saturated calomel electrode respectively.

Electrode potential, $E = E^{\circ} + \frac{2.303RT}{nF} \log \frac{1}{[Cl^-]^2}$

$$E = E^{\circ} - 0.0591 \log [Cl^-] \text{ at } 298K.$$

The electrode potential is decided by the concentration of chloride ions and the electrode is reversible with Cl^- ions.

The Emf of -

Decinormal - 0.1N soln = 0.334V

Normal - 1N KCl = 0.281V.

Saturated KCl soln = 0.2422V.

Uses :-

- It is used as a secondary reference electrode in the measurement of single electrode potential.
- It is the most commonly used reference electrode in all potentiometric determinations.

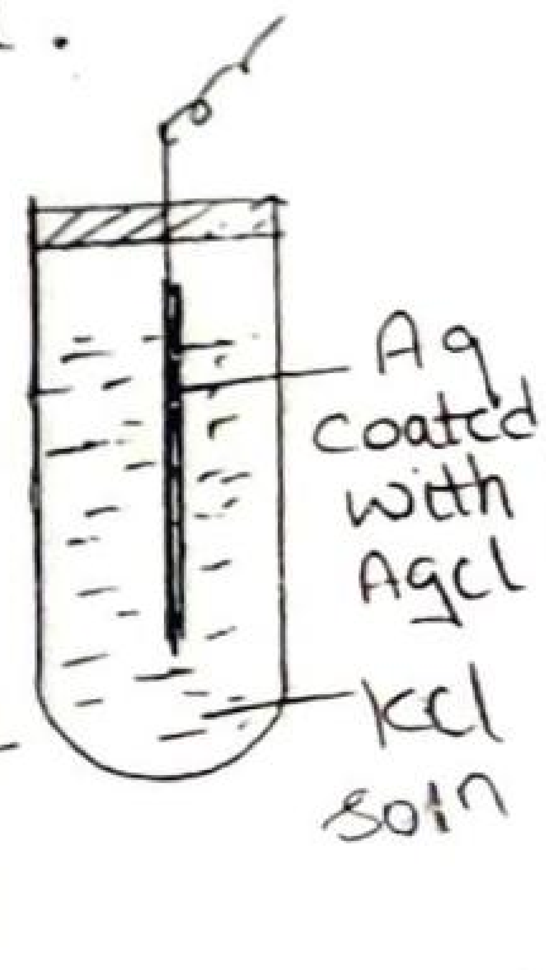
(2) silver - silver chloride electrode :-

Silver - silver chloride electrode is also a metal - metal salt ion electrode. Silver and its sparingly soluble salt silver chloride is in contact with a solution of chloride ions.

Generally a silver wire is coated with AgCl and dipped in a solution of KCl (or) HCl.

The electrode can be represented as,

$$Ag / AgCl / Cl^-$$



This electrode can also act as anode (or) cathode depending on the nature of the other electrode of the cell.

When it acts as anode.

$$Ag \longrightarrow Ag^+ + e^-$$
$$Ag^+ + Cl^- \longrightarrow AgCl$$
$$\overline{Ag + Cl^- \longrightarrow AgCl + e^-}$$

When it acts as cathode.

$$Ag^+ + e^- \longrightarrow Ag$$
$$AgCl \longrightarrow Ag^+ + Cl^-$$
$$\overline{AgCl + e^- \longrightarrow Ag + Cl^-}$$

The net electrode reaction is,

$$AgCl + e^- \rightleftharpoons Ag + Cl^-$$

Electrode potential. $E = E^\circ - 0.0591 \log [Cl^-]$.

The electrode potential is dependent on the conc. of Cl^- ions. For 1M solution, the electrode potential is 0.223V and for saturated solution it is 0.199V.

<u>Applications of Ag-AgCl electrode</u> :-

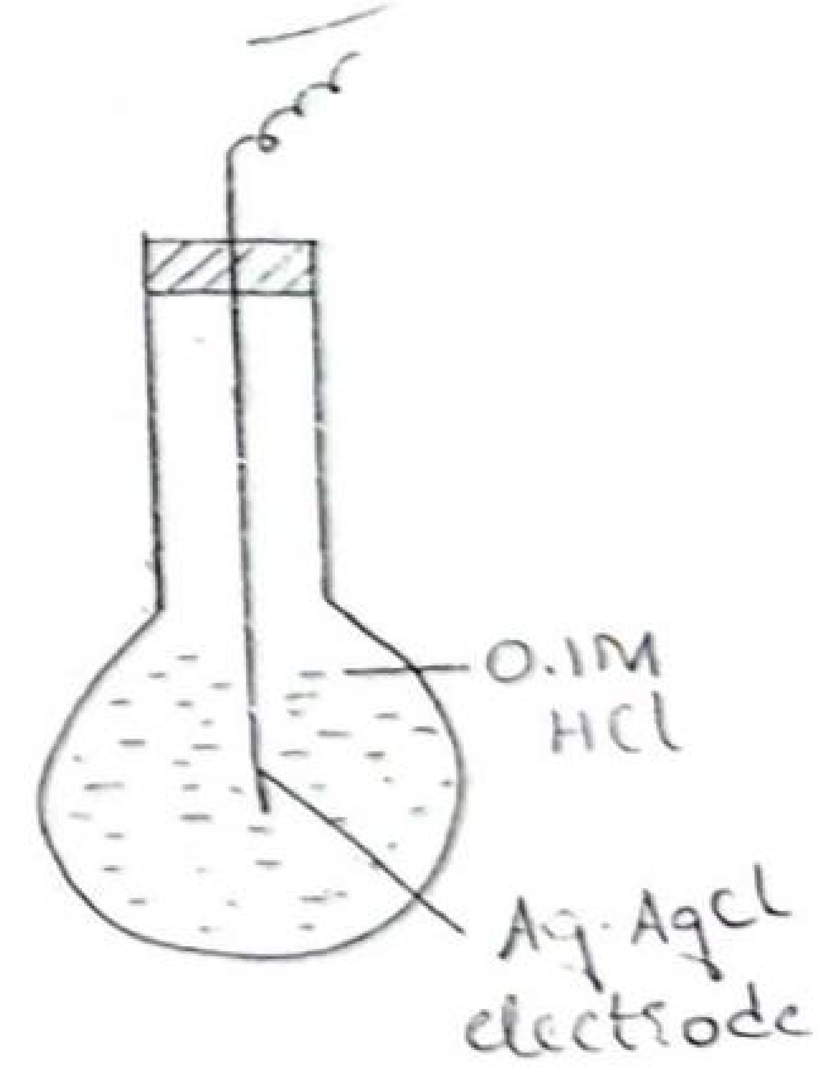
1) As a secondary reference electrode in place of calomel electrode.
2) In determining whether the potential distribution is uniform or not in ship hulls and old pipelines protected by cathodic protection.
3) As a portable reference electrode for measuring the different depths of oil rigs and submerged oil pipelines etc. Usually such probe is powered by Ni-Cd portable battery and can operate to a depth of 300-400m with precision of ±1mV.

Ion-selective electrodes :-

These are selective or responsive to a particular ion. These electrodes generally consists of a thin membrane in contact with an ionic solution.

They are so sensitive in their response that even in a solution containing small amounts of different types of ions, concentrations of particular ionic species can be measured by using them. Ion selective electrodes which are sensitive to H^+, Na^+, K^+, NH_4^+, Ca^{2+} and NO_3^- ions are available.

Glass Electrode is the most commonly used ion selective electrode.

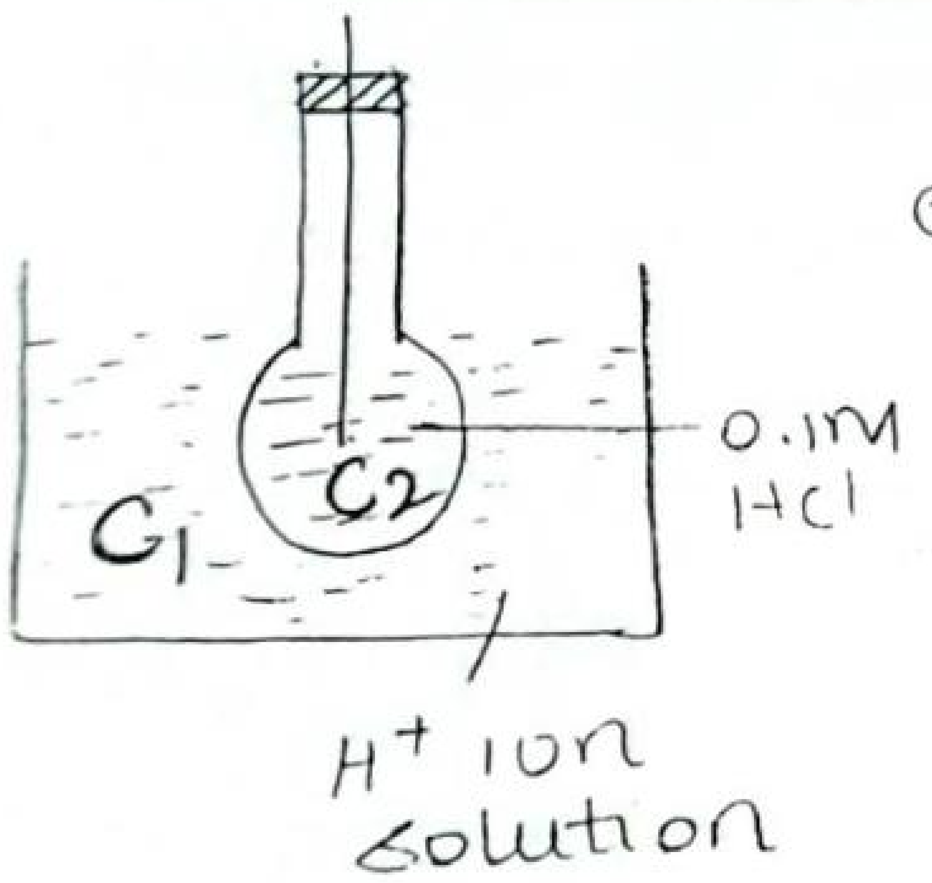


Glass electrode works on the principle that when a thin and low resistivity glass membrane is in contact with a solution, a potential is developed between the glass membrane and the solution. The potential so developed is a linear function of $[H^+]$ ions of the solution or glass electrode is responsive only to H^+ ions.

A glass electrode consists of a thin glass bulb made out of a special type of glass having low resistivity or high conductivity. The approximate composition of thin glass is 72% SiO_2, 22% Na_2O and 6% CaO. The bulb is filled with 0.1M HCl and a Pt wire or Ag-AgCl electrode is dipped into it to make electrical contact.

The bulb of the glass electrode is dipped into any solution containing H^+ ions. If the H^+ ion concentration of the solution inside and outside the glass membrane are different then a potential develops across the glass membrane.

If the conc. of H^+ ions of the solutions inside and outside the membrane are same, no potential should be developed. This potential is called the asymmetric potential.



when dipped in solution the electrode is represented as,

External solution $[H^+]=C_1$	E_1 E_2	0.1M HCl $C_2=0.1$	Ag-AgCl electrode E_{ref}

$E_b = E_1 - E_2$ $\quad$ E_b = boundary potential

The exchange of ions by the inner and outer membrane give rise to a boundary potential.

The potential of the glass electrode E_g has three components → ① the boundary potential, E_b. ② the potential of the internal Ag-AgCl reference electrode and ③ a small asymmetry potential.

$$E_g = E_b + E_{Ag} + E_{asym}$$

$$E_b = \frac{0.0591}{n} \log \frac{C_1}{C_2}$$

$$= \frac{0.0591}{n} \log C_1 \underbrace{- \frac{0.0591}{n} \log C_2}_{K}$$

$$E_b = \frac{0.0591}{n} \log C_1 + K$$

$$E_g = \frac{0.0591}{n} \log C_1 + (K + E_{Ag} + E_{asym})\ E_g^0$$

$$E_g = E_g^0 + 0.0591 \log [H^+]$$

$$E_g^0 = K + E_{Ag} + E_{asy}.$$

$$\boxed{E_g = E_g^0 - 0.0591\ pH}$$

The glass electrode is represented as,

Ag | AgCl electrode | 0.1M HCl | Glass.

Determination of pH of a given solution using glass electrode :-

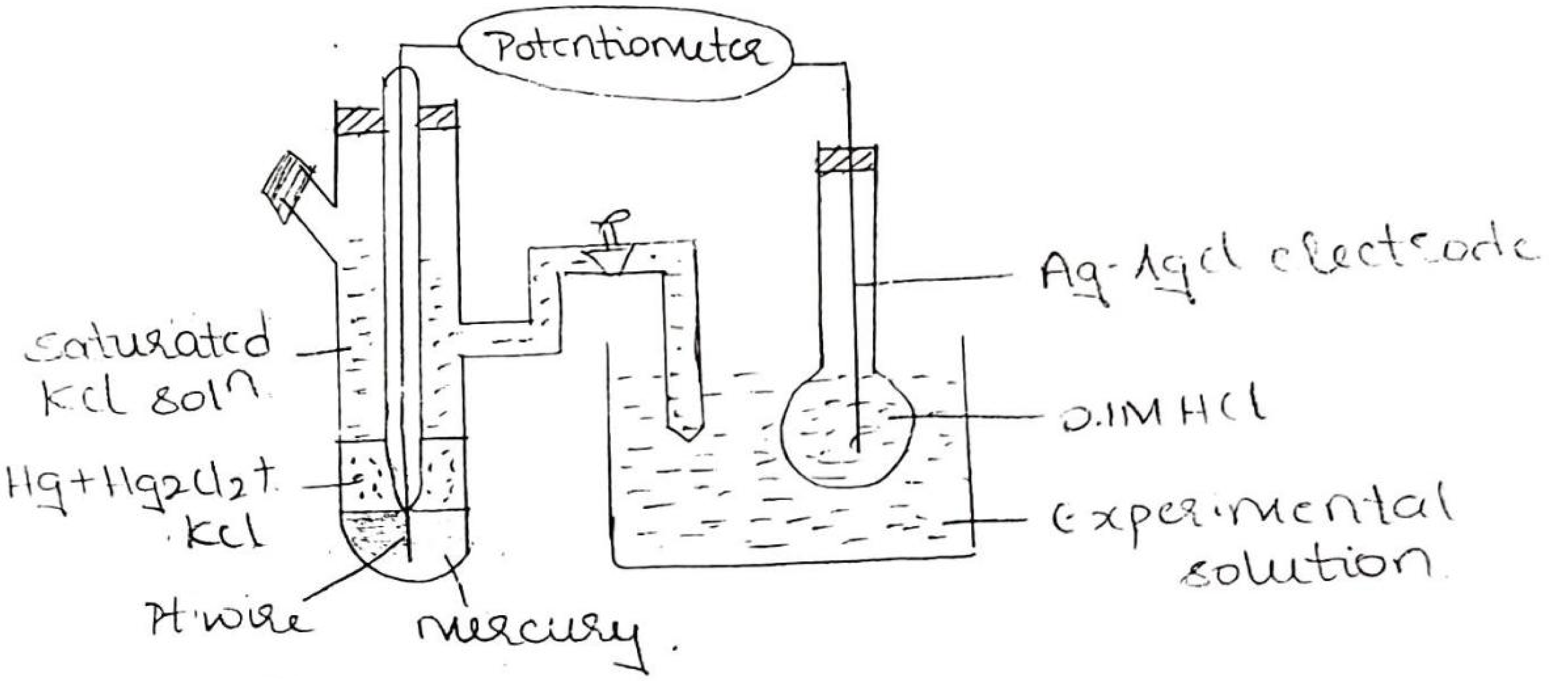
Determination of pH using glass electrode is based on the principle that when a glass membrane separates two solutions of different pH values, a potential difference is found to exist between the two surfaces of the glass which varies with the difference in the pH of the two solutions.

A glass electrode consists of a thin walled glass bulb made out of a special type of glass having relatively low melting point and high electrical conductivity. The bulb is fused to the bottom of the glass tube and is filled with 0.1M HCl. Then a pt wire @ Ag-AgCl electrode is dipped in the solution for making electrical contact. The emf of the glass electrode depends upon the pH of the solution in which it is immersed and is given by the equation.

$$E_g = E_g^0 - 0.0591\ pH$$

The glass electrode is immersed in the experimental solution and then it is combined with saturated calomel electrode (reference electrode) whose emf is 0.2422V at 25°C.

• The combination of glass electrode with saturated calomel electrode is as shown in the figure.



$Hg / Hg_2Cl_2 / Cl^- /$ Experimental solution / Glass / 0.1M HCl / Ag-AgCl electrode

The emf of the cell is determined with the help of a potentiometer & is given by,

$$E_{cell} = E_{cathode} - E_{anode}$$

$$E_{cell} = E_{glass} - E_{calomel}$$

$$E_{cell} = E_g^{\circ} - 0.0591\,pH - 0.2422$$

$$0.0591\,pH = E_g^{\circ} - E_{cell} - 0.2422$$

$$\boxed{pH = \frac{E_g^{\circ} - E_{cell} - 0.2422}{0.0591}}$$

Advantages :-

1) The electrode is simple to operate.
2) It gives accurate pH values.
3) It is not affected by any oxidizing/reducing agents present in the experimental solution. This is the major advantage of this electrode over other electrode and hence most commonly used.